

Flight Predict

Predict Departure Delay Using ML

Team 4-1

UC Berkeley School of Information



Team Members



Eric Wu



Kowsalya
Ganesan



Naresh
Kumar



Yusheng
Lee



Steven
Au

Outline

1. Abstract
2. Project Description
3. EDA
4. Feature Engineering
5. Feature Selection
6. Models
7. Conclusion



Abstract

Flight delays disrupt operations, reduce passenger satisfaction, and create system-wide inefficiencies. This project predicts **delay vs. no delay** at departure to enable proactive planning and smoother airport management.

We used 5 years of flight + weather data and engineered **50+ features** capturing congestion, weather, seasonality, disruptions, events, and airline reputation. After extensive preprocessing, we trained multiple models (XGBoost, Random Forest, MLP) to classify departure delay.

The final XGBoost model achieved an **Recall of 66%**, with scheduled departure time and recent route performance emerging as the strongest predictors. These insights support improved resource allocation, delay mitigation, and overall on-time performance.

Project Description

- **Flight delays:** ~ \$30–\$35B annually
- **Prediction:** delay vs. no delay
- **Key Metric:** Recall
- **Classification:** Risk management

- **5 years of U.S. On-Time Performance data** (31M+ records, 216 raw features)
- [ASOS weather](#)
- [flight data](#)
- [Airport data](#)
- Data Cleaning, Data Preprocessing.
- Holiday/event windows, disruptions, and airline-level metrics.

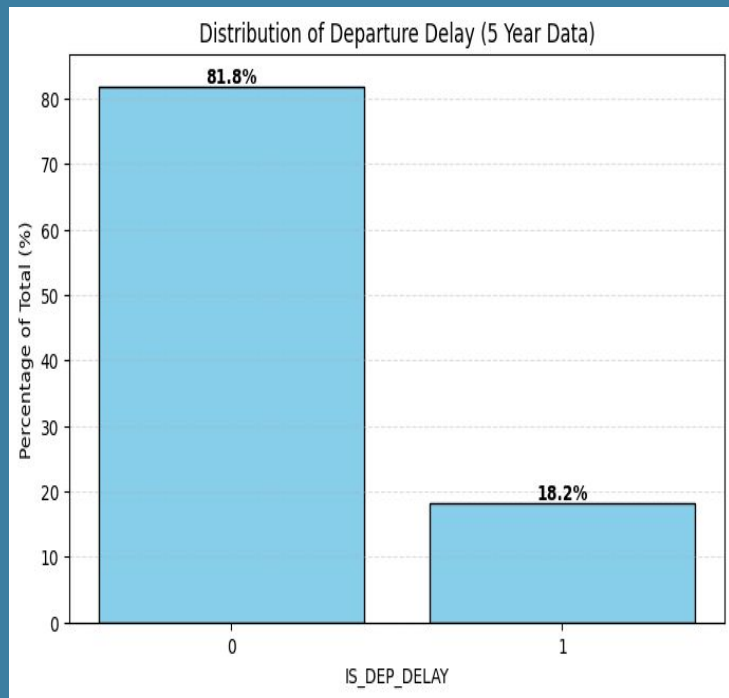
Advanced features

- Time-based signals
- Cyclical encodings
- Recent performance trends
- Graph-based airport connectivity (PageRank),
- Airline reputation

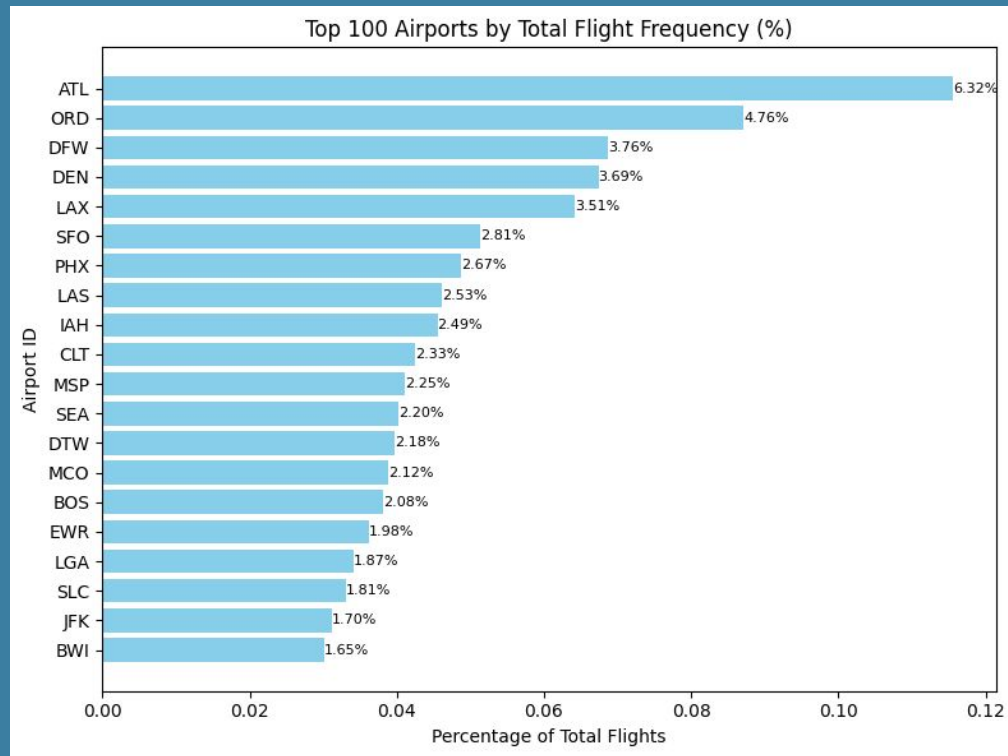
Binary classifier:

- Random Forest,
- Gradient-Boosted Trees
- MLP neural networks

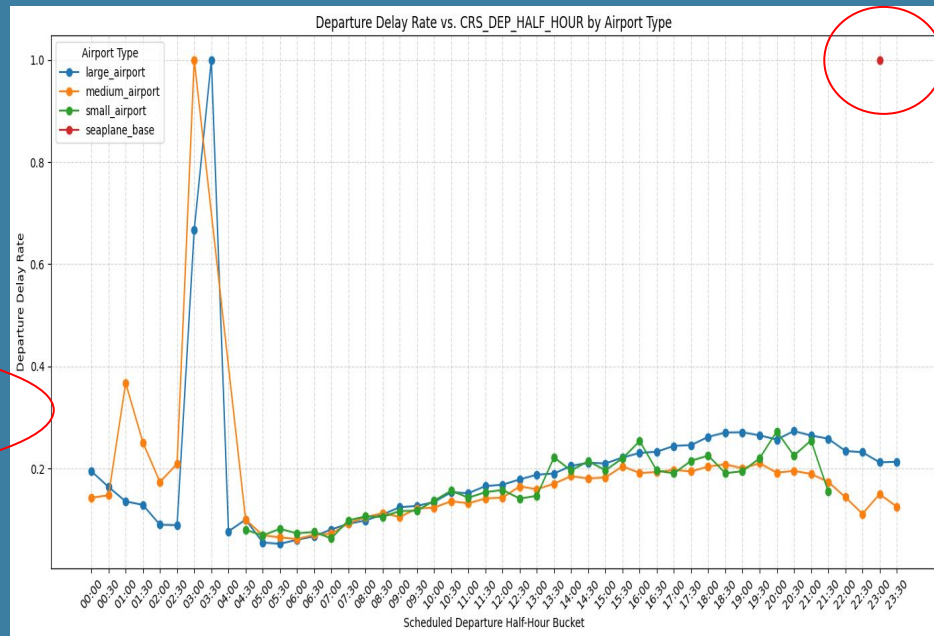
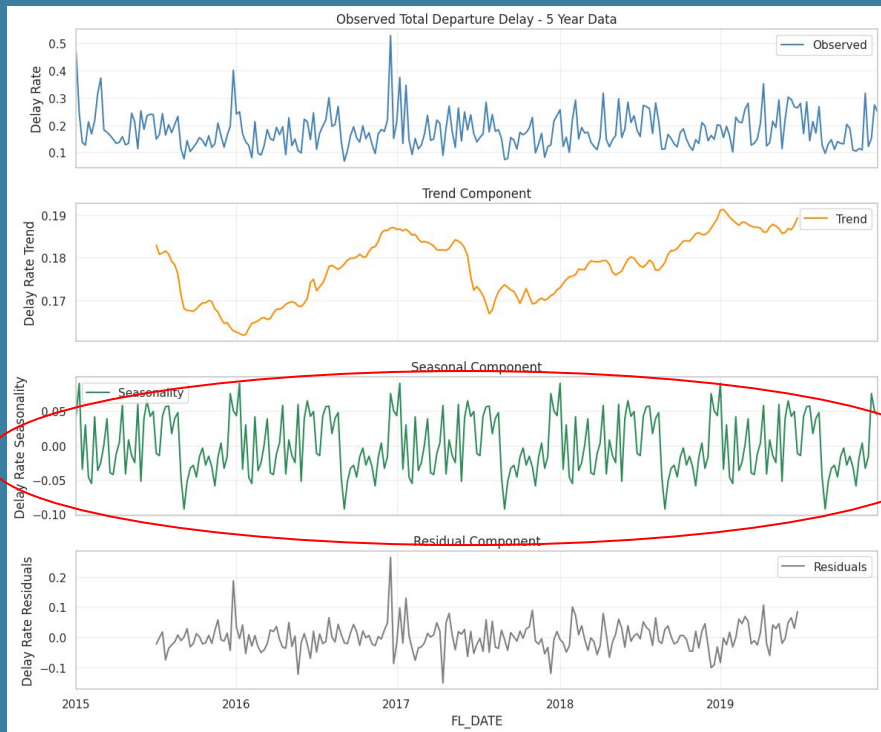
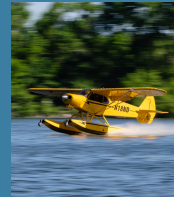
EDA Summary



Departure delays are highly imbalanced, with the majority of flights departing on time.



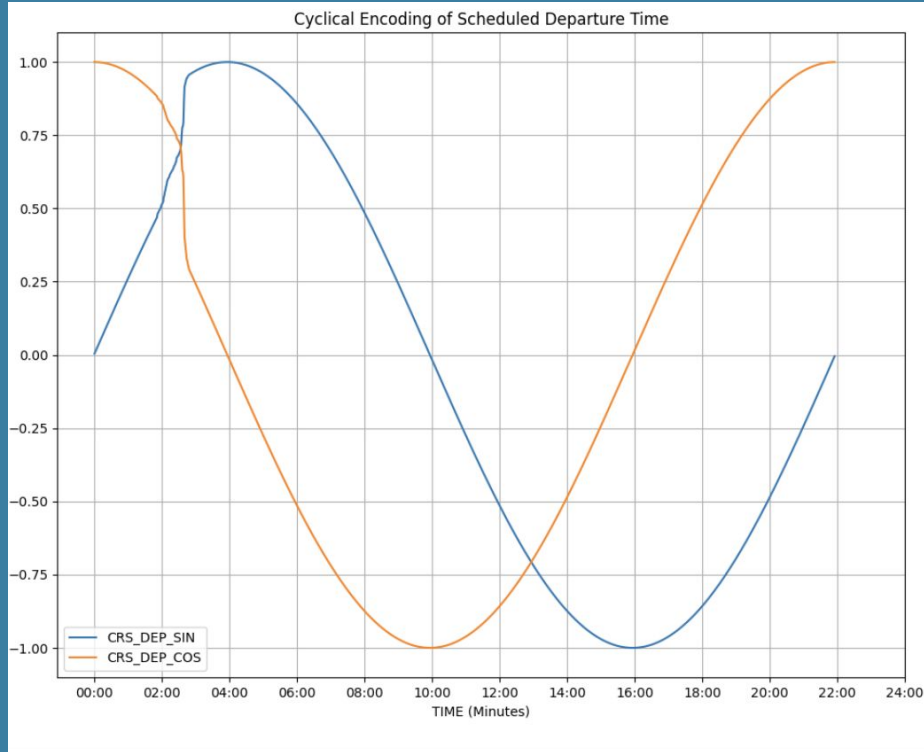
EDA Summary



- Small airport starts late
- Trend deviates later in the day
- Seaplane base outlier

Feature Engineering

Sine Cosine Encoded Departure Time

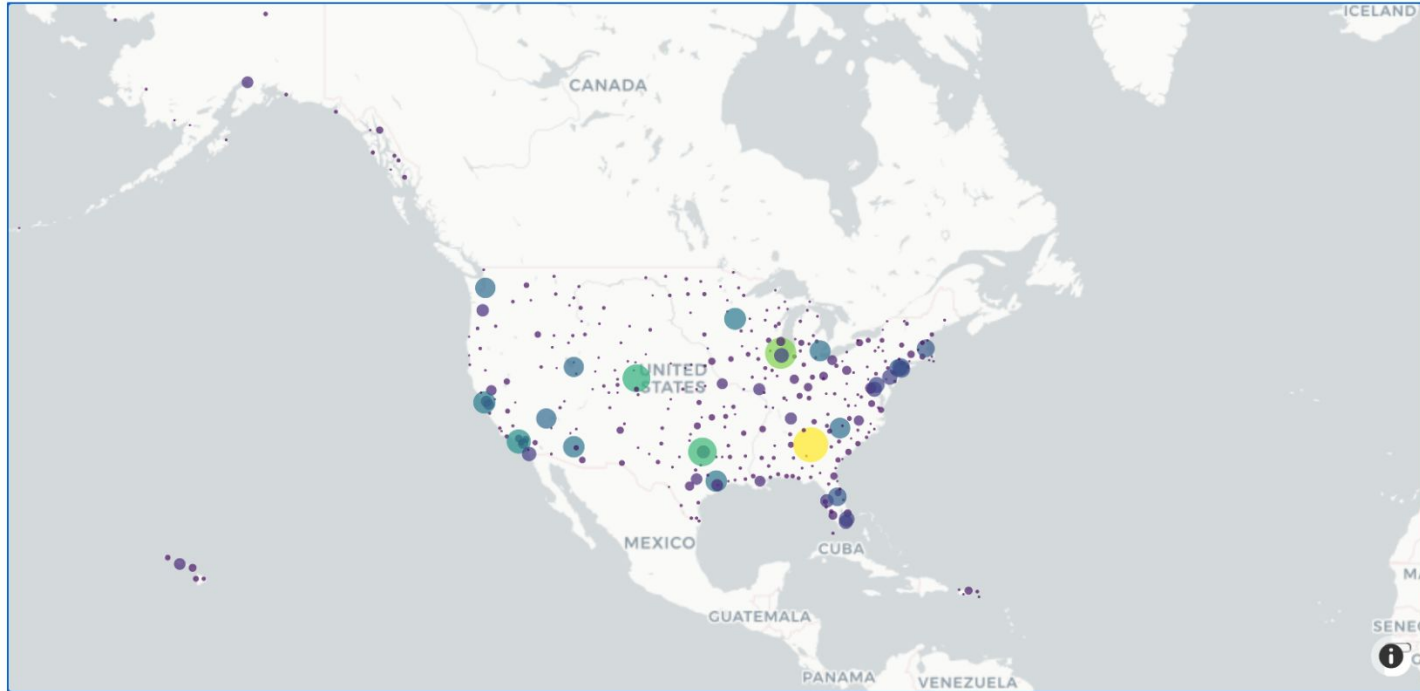


$$\sin\left(2 \cdot \pi \cdot \frac{TimeInDayInMinutes}{1440}\right)$$

$$\cos\left(2 \cdot \pi \cdot \frac{TimeInDayInMinutes}{1440}\right)$$

PageRank Airport Score

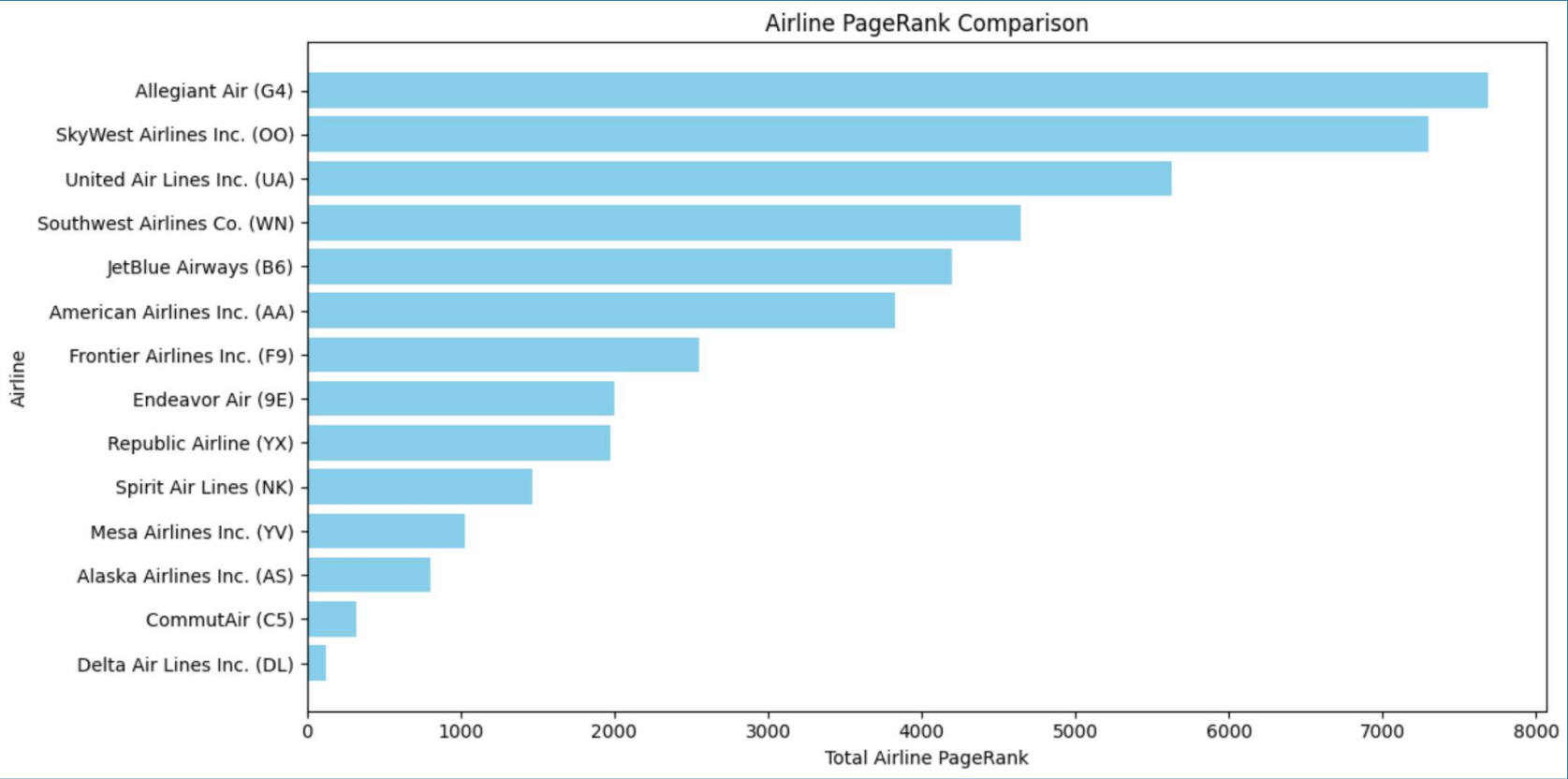
Airport PageRank Scores Weighted By Historical Flight Count



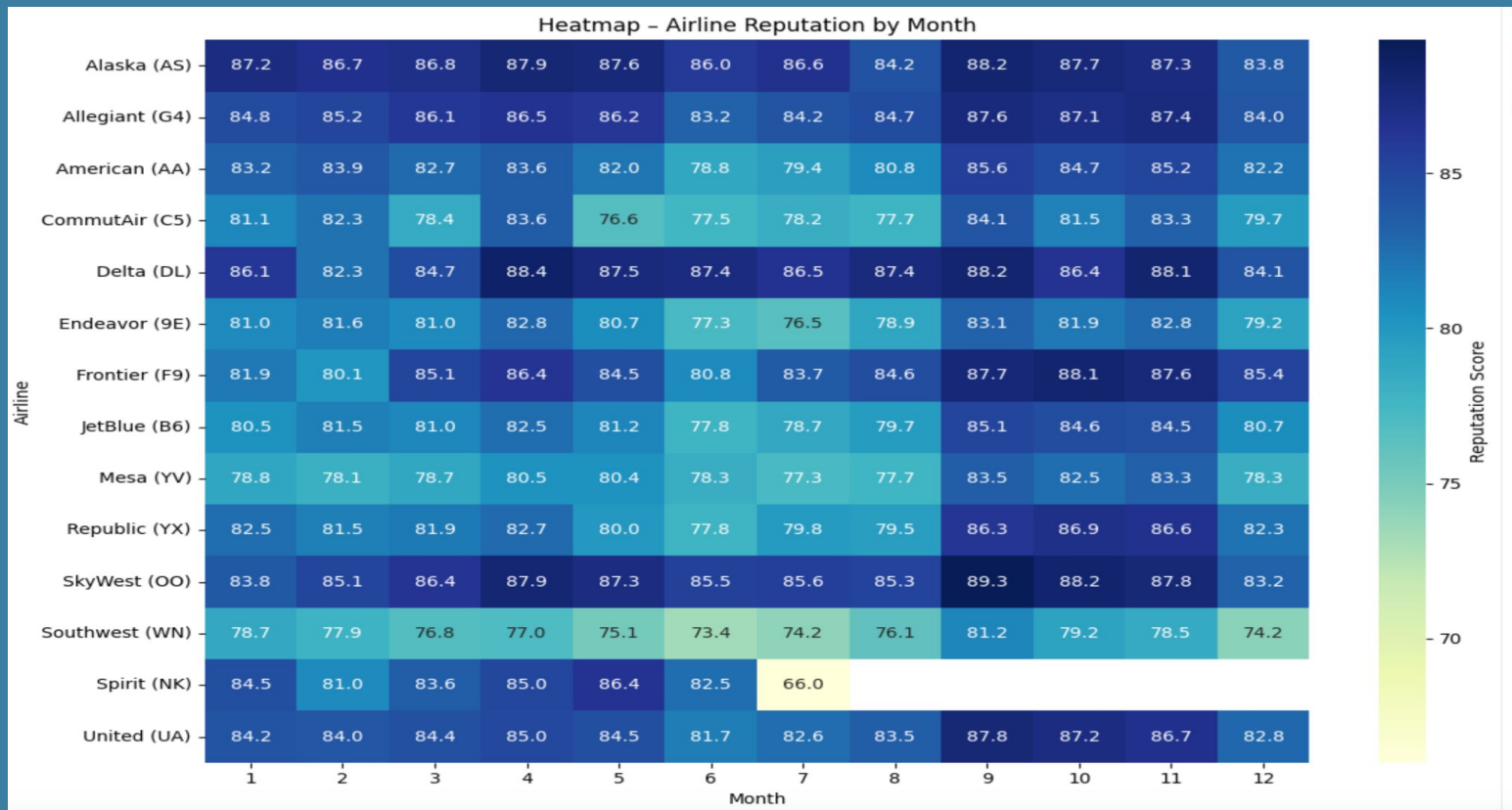
FlightCount_PR_Scores



Page Rank Airline



Airline Reputation Score



Domain-Driven and Event-Driven Features

Time-Based Features

Recent Delay Trends (7-day rolling): route_avg_delaylast_7_days, origin/dest_avg_delay_7_days

Seasonality: is_summer_peak, is_winter_peak, season

Week patterns: weekday/weekend indicators

Event-Based Features

US Federal Holidays (2015–2019)

Holiday windows (± 3 days)

Major events: Super Bowl window, CES Conference (Las Vegas)
Grace Hopper Conference

Natural Disaster Features

- disaster_severity — intensity of event
- days_since_disaster — recovery timeline
- is_disaster_day — immediate-day shock indicator

Weather-Based Features

Weather Index - PCA-derived composite score

Parsed layer of Sky Conditions to **Overall Sky Conditions** per timestamp

Airport Maintenance & Disruptions

Airport disruption flags:

- is_disruption_day
- is_airport_disrupted
- is_federal_shutdown_day

Maintenance/operational impact:

- disruption_severity
- days_into_disruption

Feature Selection

Data Integrity Checks:

Removed leakage features (e.g., wheels_off, arrival timestamps) that reveal information *after* departure.

Redundancy Check (Pearson Correlation):

Dropped one feature from each pair with $r > 0.95$ to avoid multicollinearity.

Statistical Significance (ANOVA F-test):

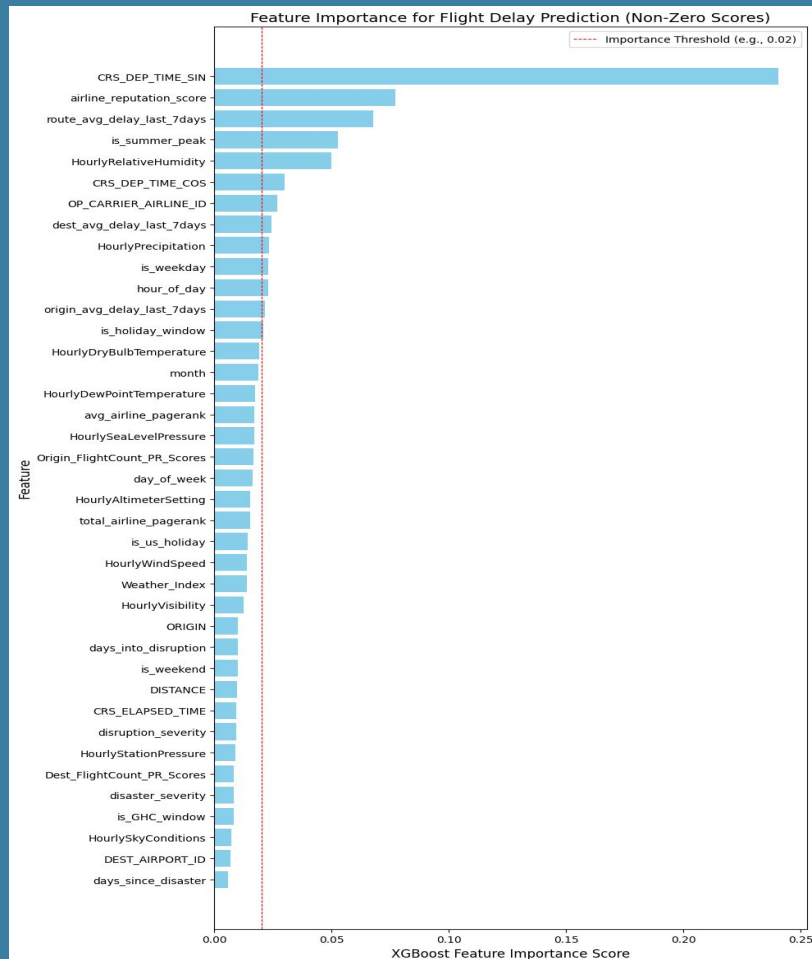
Removed features with **low F-Statistic** or **P-Value** > 0.05 , indicating weak predictive power.

XGBoost Feature Importance:

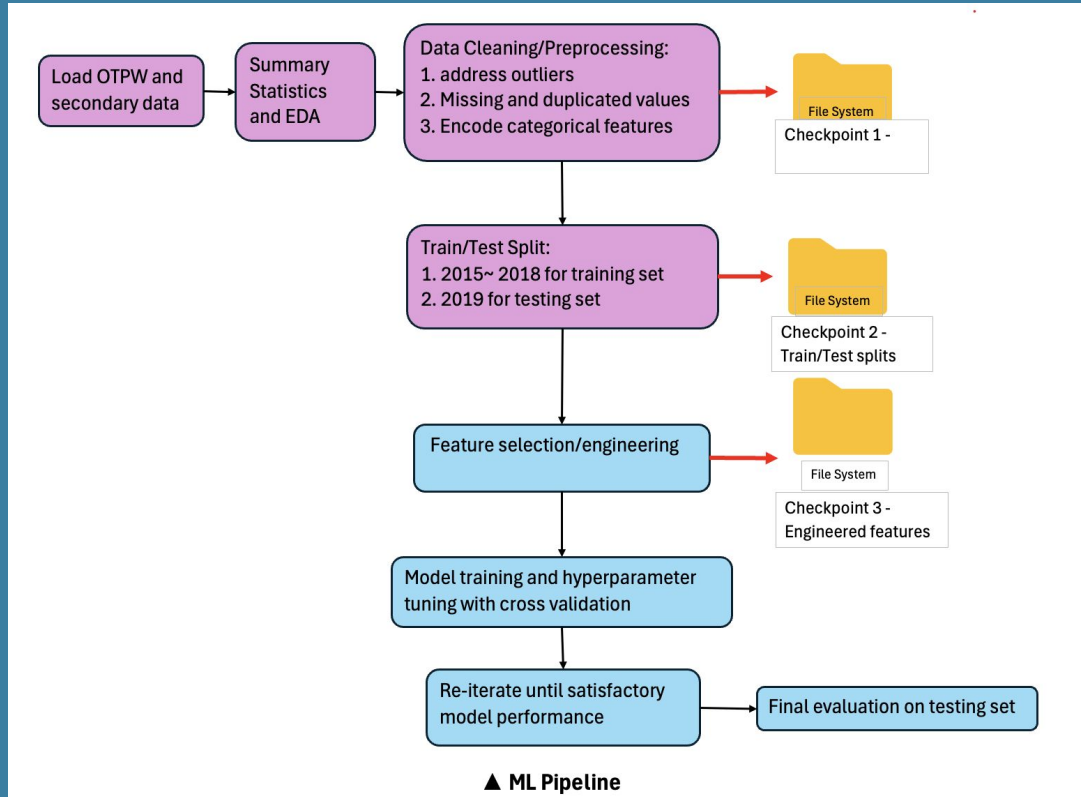
Identified and retained the **38 features** with non-zero importance for model training.

Top Predictive Features:

- CRS_DEP_TIME_SIN (departure time cyclical encoding)
- Airline reputation score
- Route-level delay trend (7-day rolling)



Model pipeline



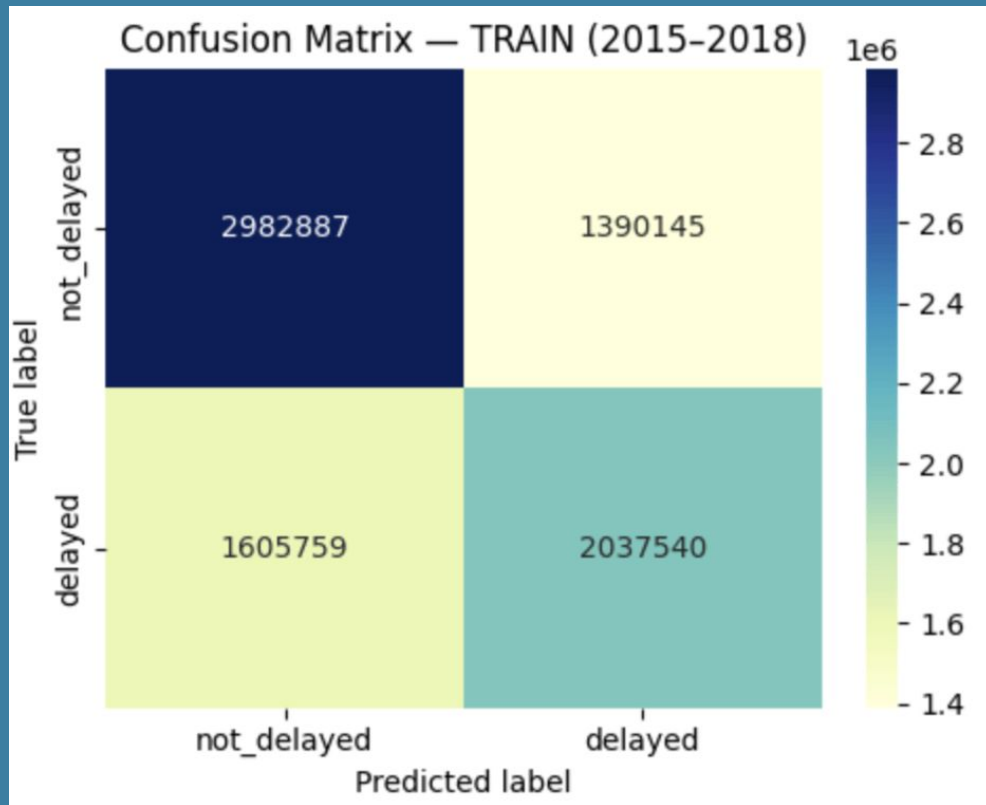
Multilayer Perceptron - MLP (5 year)

Model tuning(Grid search):

- hidden_layers
- downsampling_ratio
- threshold

Best Parameters:

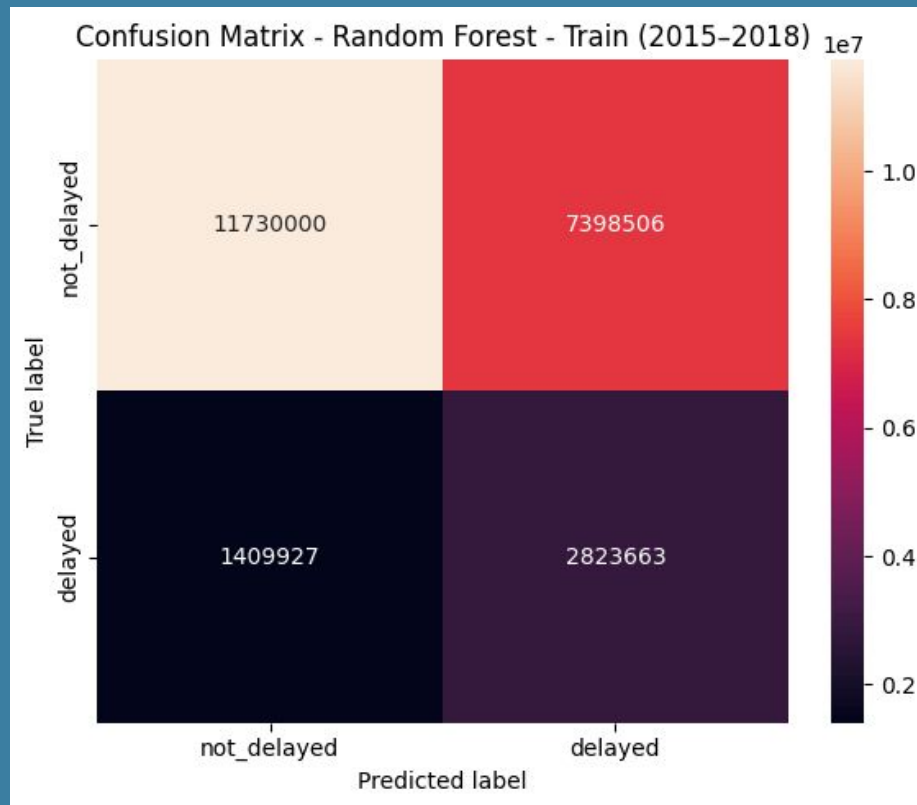
- hidden_layers = [64, 32]
- downsampling_ratio = 1.2
- threshold = 0.5



Ensemble Decision Tree: Random Forest (5 year)

Best Parameters (Grid Search):

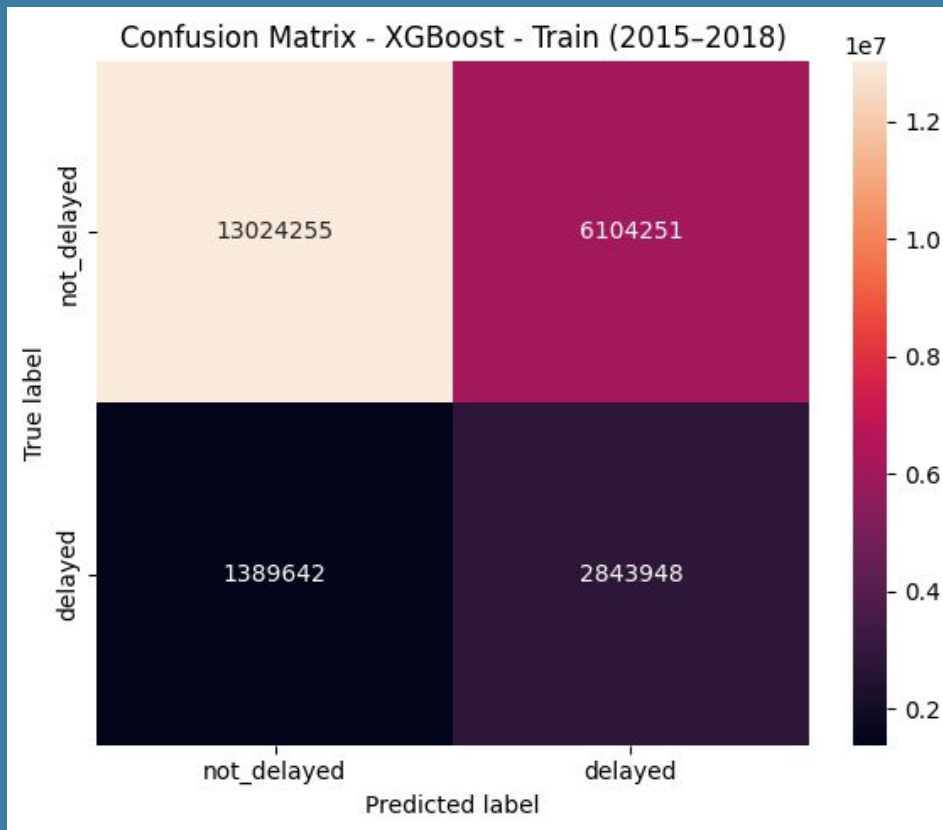
- numTrees = 100
- maxDepth = 8
- maxBins = 32
- featureSubsetStrategy = auto



Gradient Boosted DT Model: XGBoost (5 year)

Best params (Grid Search):

- Max_depth = 10
- n_estimators = 175
- learning_rate = 0.1
- subsample = 0.8
- colsample_bytree = 1.0
- reg_lambda = 2.0
- reg_alpha = 1.0

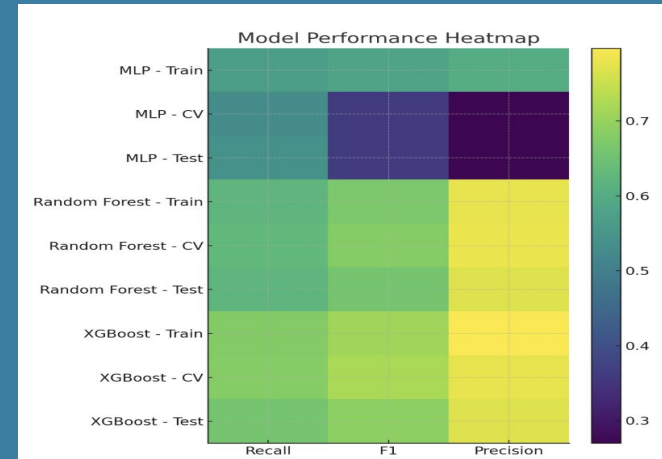
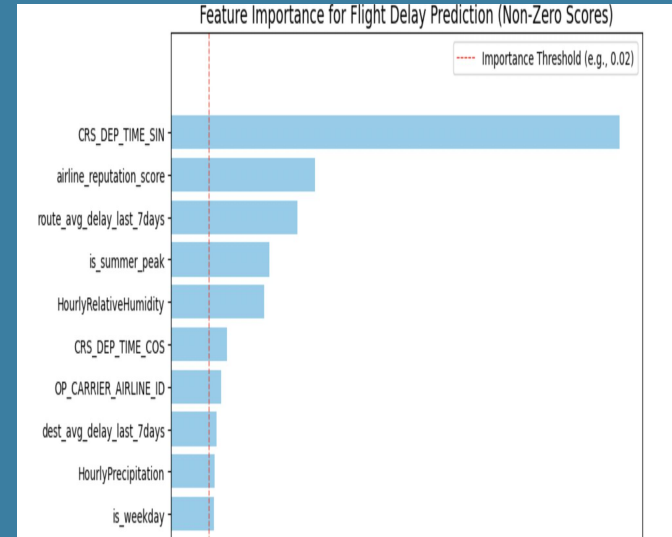


Model Performance Comparison

Model	Stage	Recall	F1	Precision
MLP	Train	0.57	0.58	0.60
	CV	0.53	0.36	0.27
	Test	0.54	0.36	0.27
Random Forest	Train	0.622	0.67	0.78
	CV	0.63	0.68	0.781
	Test	0.62	0.66	0.77
XGBoost	Train	0.679	0.71	0.797
	CV	0.68	0.72	0.78
	Test	0.66	0.69	0.77

Conclusion

- Feature rich approach
- Key Predictive Drivers
- Optimized modeling & tuning
- Best overall model
- Next steps



Thank You !

Icon Library

Use these icons to represent different career paths, disciplines, and technologies.



Machine Learning



Coding



Cybersecurity



UX Design



Data Science



HCI



Business



Law



Library Studies



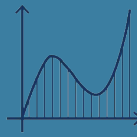
Ethics



Economics



Research



Math



Data Visualization



Program Manager



Psychology